

Prediction Markets as DeGroot Dynamics

with Time-Varying Communication Topology

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Guest Lecture · Random Signal Processing

Why Prediction Markets?

The Core Question

How do decentralized agents, each with partial and noisy information, aggregate knowledge into a single price that converges to truth?

Key Insight

A prediction market is an implicit DeGroot consensus process. Traders never talk directly, the market price encodes past beliefs, creating peer influence through the price channel.

Signal Processing Angle

Noisy signals $\rightarrow$ belief updates $\rightarrow$ price dynamics.

The market acts as a nonlinear filter fusing correlated, biased random signals into an estimate of a hidden binary state θ .

Model Setup

$\theta \in \{0, 1\}$ — binary ground truth

$K = 4$ information sources with bias μ_j

$N = 6$ traders with heterogeneous attention

Each source emits:

$$s_j(t) = \theta + \mu_j + \varepsilon_j(t)$$

$$\varepsilon_j \sim N(0, \sigma^2)$$

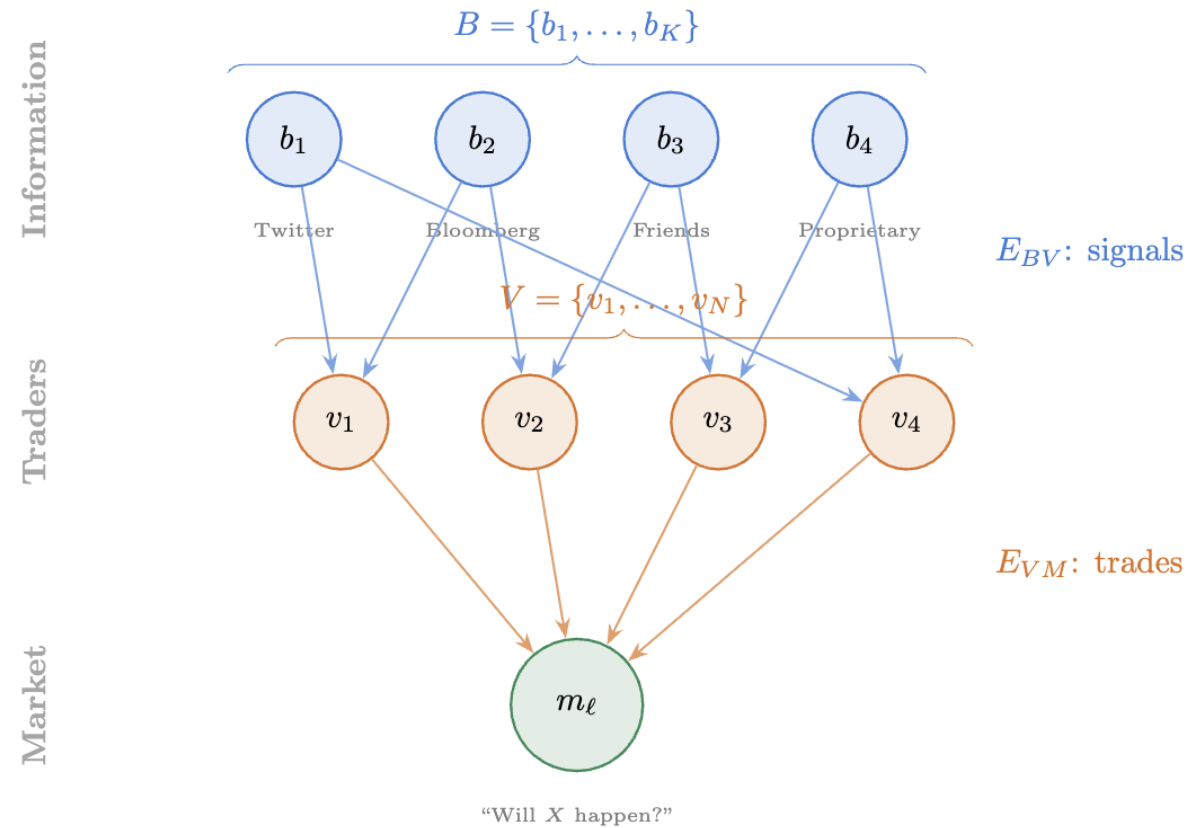
Three-layer architecture:

Sources (B) $\rightarrow$ Traders (V) $\rightleftharpoons$ Market (M)

Topology grows over time as traders enter and trade, adding edges to $D(t)$.

Model Formulation

1.1 The Three Layers



Model Formulation: Information Sources

4.1 Information Sources (Stubborn Agents)

There are K information sources. Each source b_j emits a signal at each time step:

$$s_j(t) = \theta + \mu_j + \varepsilon_j(t), \tag{1}$$

Model Formulation: Information Sources

$$\theta \in \{0, 1\}$$

$\mu_j \in \mathbb{R}$ is a *persistent bias*

$\varepsilon_j(t)$ is zero-mean noise

$$\text{Cov}(\varepsilon_j, \varepsilon_k) = [\Sigma_B]_{jk}$$

Model Formulation: Trader Belief

$$p_i(t) = \underbrace{\beta_i^{\text{info}} \cdot \sigma(\mathbf{w}_i^\top \mathbf{s}(t))}_{\text{information from sources}} + \underbrace{\beta_i^{\text{price}} \cdot \pi_\ell(t)}_{\text{information from price}} + \underbrace{(1 - \beta_i^{\text{info}} - \beta_i^{\text{price}}) \cdot p_i(t^-)}_{\text{prior belief (inertia)}}$$

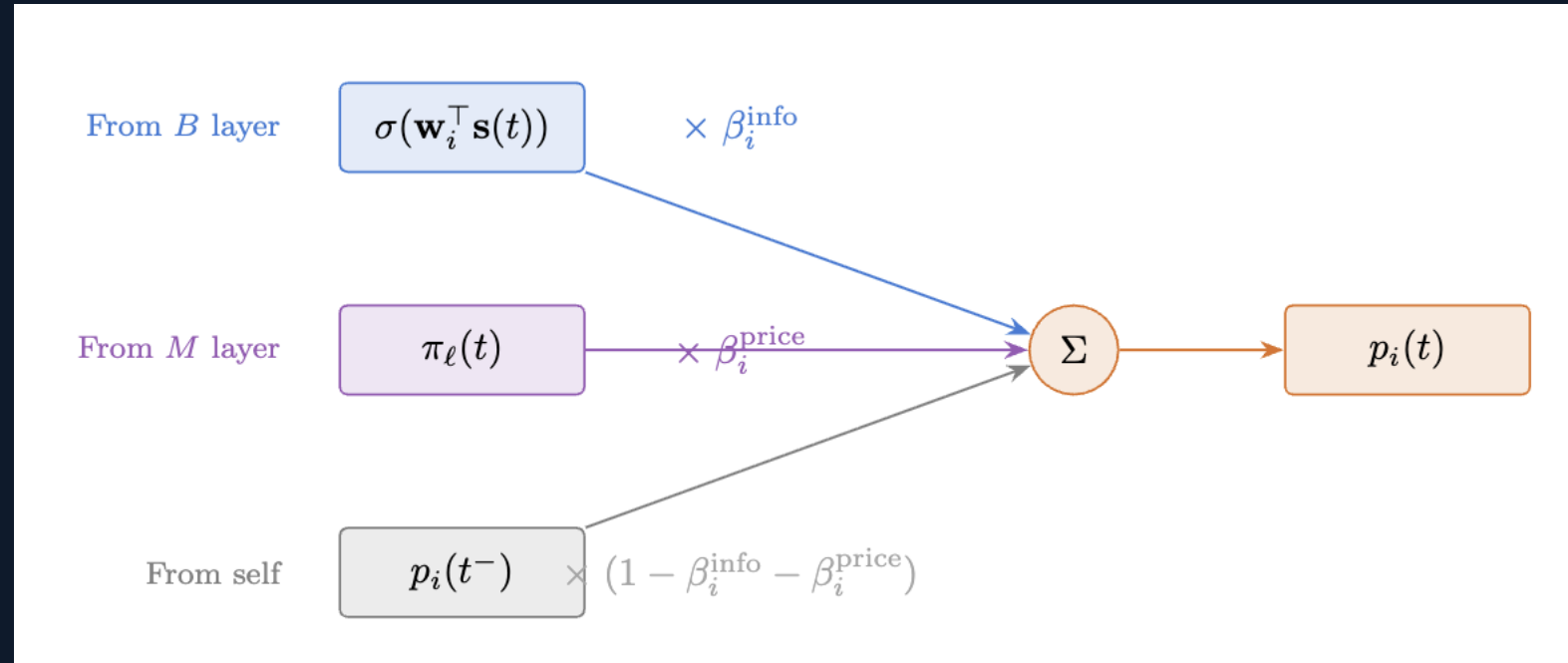
Model Formulation: Trader Belief

4.2 Traders (Regular Agents)

There are N traders. Each trader v_i is parameterized by:

Symbol	Name	Meaning
$\mathbf{w}_i \in \mathbb{R}_{\geq 0}^K$	Attention vector	How much trader i attends to each source. Entry w_{ij} is the weight on source b_j . This defines the $B \rightarrow V$ edges.
$a_i > 0$	Risk aversion	Scales position size inversely: more risk-averse traders bet smaller.
$\lambda_i > 0$	Arrival rate	Rate of a Poisson process governing when trader i checks the market.
$\alpha_i \in (0, 0.5]$	Deadband	Trader i only bets when $ p_i(t) - \pi(t) > \alpha_i$ (minimum perceived edge).
$\beta_i^{\text{info}} \in [0, 1]$	Information weight	How much trader i trusts their own information sources.
$\beta_i^{\text{price}} \in [0, 1]$	Price weight	How much trader i trusts the market price as a signal.

Model Formulation: Trader Belief



Model Formulation: From Belief to Trade

6.1 Demand Function

The trader's demand (signed bet quantity) is:

$$q_i(t) = \begin{cases} \frac{p_i(t) - \pi_\ell(t)}{a_i}, & \text{if } |p_i(t) - \pi_\ell(t)| > \alpha_i, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The deadband α_i ensures traders only bet when their perceived edge exceeds a threshold.

Model Formulation: From Belief to Trade

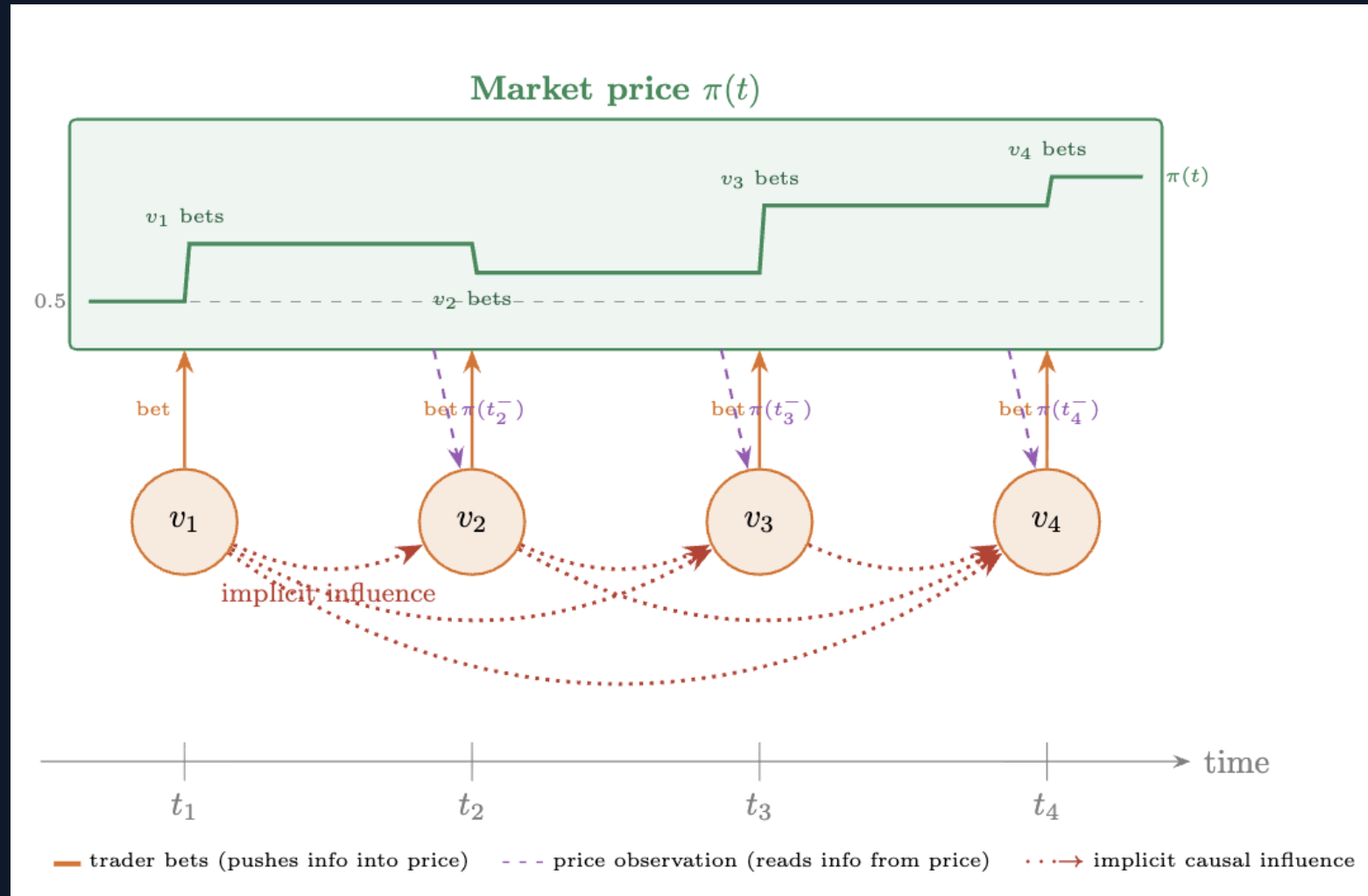
6.2 Price Impact

When trader i bets, the market maker absorbs the order and adjusts the price:

$$\pi_\ell(t^+) = \text{clip}_{[0,1]} \left(\pi_\ell(t) + \gamma_\ell \cdot q_i(t) \right). \quad (5)$$

γ_ℓ controls how much a single trade moves the price.

Model Formulation: From Belief to Trade

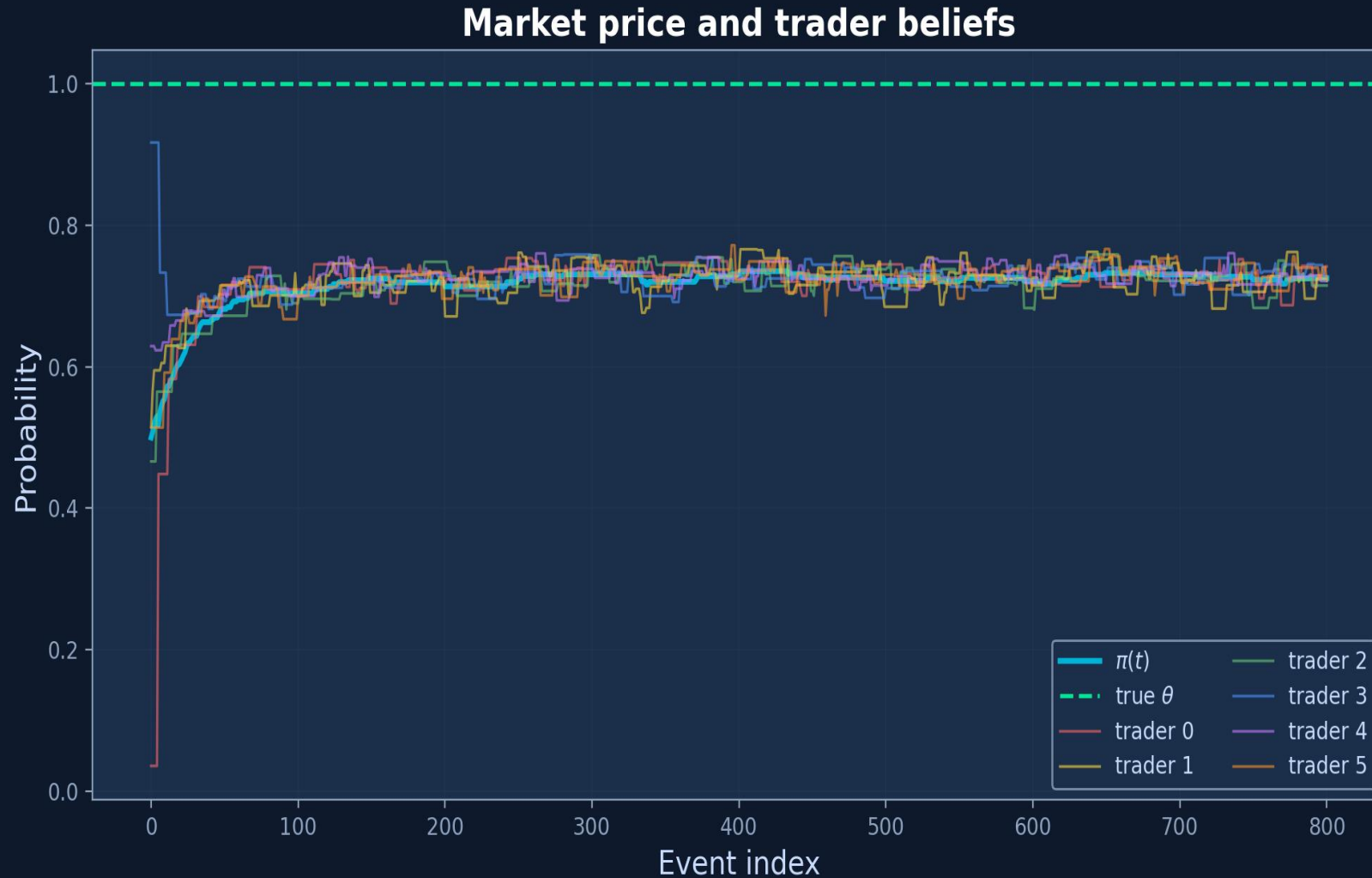


Model Formulation: From Belief to Trade

Key Insight

Even though there are no explicit $V \rightarrow V$ edges in the tripartite graph, the price feedback channel creates **implicit causal influence** between traders. The influence is *directed* (only past bets affect future traders) and *time-varying* (the set of influential traders grows as more people bet). This is the observation that motivates the DeGroot reformulation.

Simulation: Price Convergence



Observations

800 discrete time steps
 $\theta = 1$ (green dashed line)

Three phases:

1. Seeding — sparse trades, volatile price
2. Discovery — active trading, oscillation around truth
3. Consensus — beliefs and price stabilize near θ

$\pi(t)$ (cyan) tracks the belief centroid and converges to θ .

DeGroot Dynamics & Peer Influence

The DeGroot Connection

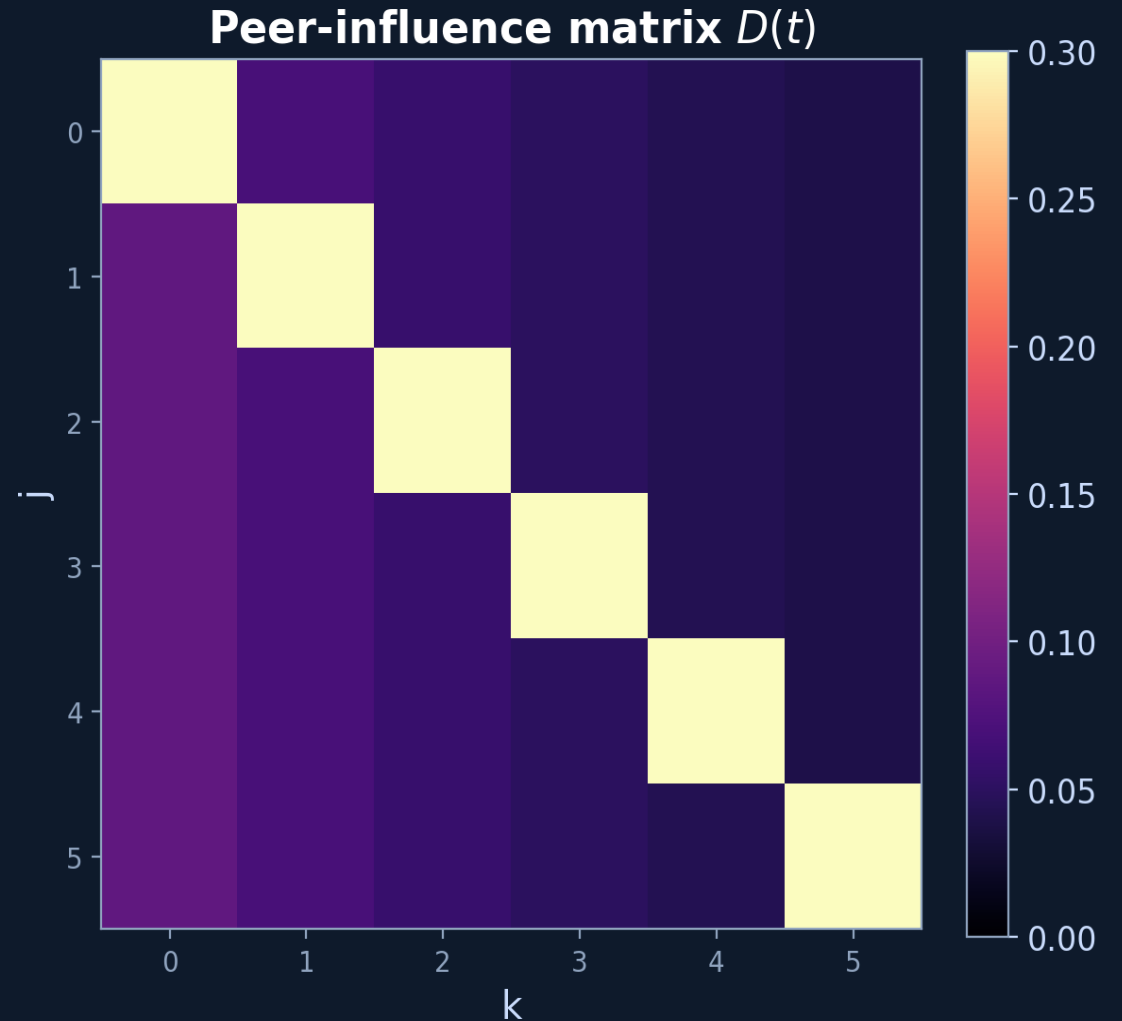
In log-odds coordinates, the belief update becomes exactly linear — a DeGroot model:

$D(t)$ is the peer-influence matrix — it grows as traders enter the market.

Time-Varying Topology

$D(t)$ starts nearly diagonal (self-weights only). As traders trade, off-diagonal entries grow — reflecting implicit influence through the price channel.

Convergence speed $\rightarrow$ spectral radius $\rho(D)$



Key Takeaways

Market as Filter

The prediction market acts as a nonlinear recursive filter — aggregating noisy, biased signals into a consensus estimate of the hidden state θ .

Connects information theory to market microstructure.

Implicit Communication

Traders never communicate directly. Yet the market price creates causal peer-to-peer influence — captured by the growing matrix $D(t)$.

Price is the communication channel in this network.

Spectral Convergence

Convergence rate depends on spectral radius $\rho(D)$. Information diversity (rank of Φ) determines whether consensus lands near θ or a biased value.

Classic DeGroot theory applies in log-odds coordinates.

Thank you · Questions?